

HW

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$\{u_1, u_2, u_3, u_4\}$ Basis

$$u_1 = \begin{bmatrix} 0 \\ 1 \\ -4 \\ -1 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 3 \\ 5 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ -4 \end{bmatrix}$$

$$u_4 = \begin{bmatrix} 15 \\ -3 \\ -1 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 10 \\ -8 \\ 2 \\ 0 \end{bmatrix}$$

$$x = z_1 + z_2$$

$$z_1 \in K^n = \text{span}\{u_1, \dots, u_4\}$$

$$z_2 \in K^{n-1} = \text{span}\{u_3\}$$

$$z_1 = c_1 u_1 + c_2 u_2 + c_3 u_3 + c_4 u_4$$

$$z_1 =$$

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{10 - 8 - 4 + 1}{1 + 16 + 1 + 1} = \frac{-1}{18} = -\frac{1}{18}$$

$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{30 - 40 + 2}{9 + 25 + 2} = \frac{-8}{56} = -\frac{1}{7}$$

$$c_3 = \frac{x \cdot u_3}{u_3 \cdot u_3} = \frac{10 + 2}{1 + 1 + 16} = \frac{12}{18} = \frac{2}{3}$$

$$c_4 = \frac{x \cdot u_4}{u_4 \cdot u_4} = \frac{50 + 24 - 2}{25 + 9 + 2} = \frac{72}{36} = 2$$

$$z_1 = -\frac{1}{18} \begin{bmatrix} 0 \\ 1 \\ -4 \\ -1 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 3 \\ 5 \\ 1 \\ 1 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} -1 \\ 0 \\ 1 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 15 \\ -3 \\ -1 \\ 1 \end{bmatrix}$$

$$z_1 =$$

$$= \begin{bmatrix} 0 \\ -\frac{1}{18} \\ -\frac{4}{18} \\ \frac{1}{18} \end{bmatrix} + \begin{bmatrix} 2 \\ \frac{10}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix} + \begin{bmatrix} -\frac{2}{3} \\ 0 \\ \frac{2}{3} \\ -\frac{8}{3} \end{bmatrix} + \begin{bmatrix} 30 \\ -6 \\ -2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 30 \\ -6 \\ -2 \\ 2 \end{bmatrix}$$

③

$$y = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$u_1 \cdot u_2 = -1 + 1 = 0 \quad \checkmark$$

$$y = \cancel{y} + z$$

$$\cancel{z} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

$$y = \sum \frac{y \cdot u_i}{u_i \cdot u_i} u_i$$

$$y = \frac{-1+4}{1+1} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \frac{1+4}{1+1} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{3}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \frac{5}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{5}{2} \\ \frac{5}{2} \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

④

$$\cancel{y} = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}$$

$$u_1 \cdot u_2 = 5$$

$$y = z_1 + z_2$$

$$z_1 = c_1 u_1 + c_2 u_2$$

$$z_2 = c_3 u_3$$

$$z_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} u_2 = \frac{10}{10} u_1 + \frac{5}{5} u_2 = u_1 + u_2$$

$$= \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix}$$

$$= 0 + \frac{28}{4} u_2 = 7 u_2$$

$$y = \begin{bmatrix} 10/3 \\ 8/3 \\ 4/3 \end{bmatrix} + \begin{bmatrix} -4/3 \\ 4/3 \\ 4/3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

$$y = z_1 + z_2$$

$$z_2 = y - z_1$$

$$= \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ 2 \end{bmatrix}$$

① closest point to y in $W = \text{span}\{v_1, v_2\}$

$$y = \begin{bmatrix} 3 \\ 1 \\ 5 \\ 1 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\hat{y} = \frac{y \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{y \cdot v_2}{v_2 \cdot v_2} v_2 = \frac{1}{2} v_1 + \frac{3}{2} v_2$$

$$\hat{y} = v_1 + \frac{3}{2} v_2 = \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$